

Chest Pain / STEMI

History / Signs and Symptoms

- Onset – When? Activity at onset?
- Chest pain – Dull? Pressure? Sharp? Area? Radiation?
- Associated symptoms – Nausea? Shortness of breath? Syncope/near-syncope? Palpitations? Abdominal pain? Anxiety?
- Past medical history/cardiac risk factors – Previous MI? Known cardiac disease? Diabetes? Hypertension? High cholesterol? Smoker? Obesity? Family history? Cocaine use?
- PE risk factors – Sedentary? Post-surgery? Cancer? Birth control use? Pregnant? Recent long bone fracture?
- Signs of pneumonia / infection – Fever? Cough?

Physical Exam

- General appearance – is patient in distress?
- Vital signs – Hypertension? Hypotension? Tachycardia? Bradycardia? Tachypnea? Hypoxia (pulmonary edema)? Fever?
- Breath sounds – Clear or wet (crackles/rales)? Diminished or absent on one side? Wheezing?
- Skin – Diaphoretic? Gray? Hot? Red?
- Mental Status – Altered? Agitated?
- EKG – Rate? Rhythm? ST Elevation greater than 1 box in two or more contiguous leads (STEMI)?
- Abdominal exam – Epigastric tenderness? Pulsatile mass?

Potentially life-threatening causes of chest pain: Myocardial Infarction (MI), Unstable Angina, Pulmonary Embolus (PE), Pneumonia, Thoracic or Abdominal Aortic Dissection, Asthma/COPD exacerbation, Mediastinitis (infection in area around heart and great vessels), Esophageal Rupture, Pneumothorax

EMT/AEMT

- Follow General Medical Care Guideline
- Follow Adult Resuscitation Guideline, as indicated
- **Aspirin** 324 mg PO, (only if No allergy, No signs of stroke and No active GI bleeding)
- Patient may take their prescribed Nitroglycerin if they are requesting to do so and SBP is > 100 systolic.
 - EMT/AEMT should not recommend Nitroglycerin without advanced level provider available

EMT/AEMT - Expanded Scope

- **NS/LR** 500ml IO x 1 bolus for SBP $\leq$ 90
- **Zofran** 4mg IM/ODT for nausea/vomiting
 - May repeat x once
 - Max 8mg total

Paramedic

Chest Pain / STEMI

- **Aspirin** 324 mg PO, if not given previously.
- 12 – Lead EKG (if available)
 - Right-sided V4R for Inferior STEMI
 - Posterior Lead for ST depression in V1
- **NS/LR** 500 ml – 2000mL IO/IV for SBP $\leq$ 90
- **Zofran** 4mg IM/IV/IO/ODT for nausea/vomiting
 - May repeat x once
 - Max 8mg total
- **Nitroglycerin** 0.4 mg SL every 5 minutes until
 - patient is chest pain free OR
 - SBP <100

Right-sided (Inferior STEMI or ST elevation in V4R)

- Hold NTG
- **NS/LR** 500 mL – 2000 mL IO/IV for SBP < 100
- Pain management per Pain Management Guideline (AVOID ketorolac/ibuprofen in STEMI and chest pain concerning for cardiac involvement)

